

Additional Experimental Results

Leduc and MetaDrive

23 September 2026

This is a standalone preview of additive project files. It does not replace or modify the existing manuscript. The Leduc second cohort and the MetaDrive first cohort are reported separately; no result from the unfinished MetaDrive precision follow-up is included. Text is a draft for author review.

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1 A second Leduc cohort: allocation, stopping, and query-size ablations

Cohort identity and protocol. We evaluated a second, pre-registered cohort designed to make acquisition rules distinguishable after an earlier cohort produced identical EVSI, IVR, and LUCB decisions. This is an outcome-informed subsequent cohort, rather than an untouched first attempt. The eight-candidate family, query menu, budgets, and analysis were fixed before generating this cohort. Four pilot roots were included in 12 calibration roots, followed by 30 separate confirmation roots (15 equilibrator and 15 exploiter worlds; seeds 98000–98029). Opponent adaptation lengths were 1 and 8. The primary long-adaptation comparison used a cap of 16,384 HF hands. A paired query of size $n \in \{512, 2048, 4096\}$ costs $2n$ hands; fixed-size comparators use $n = 4096$. Selection methods were replayed on the same sealed banks, and deployment regret was scored against exact OpenSpiel expected returns, including the incumbent as a no-update option. PIVOT-KG denotes the joint-Gaussian knowledge-gradient implementation used in this cohort; these numbers are not results for a separately implemented author baseline.

Mechanism and allocation. The mean root-level reversal rate among proxy-positive candidates increased from 2.50% under short adaptation to 51.67% under long adaptation (Figure 3). At the primary budget, PIVOT-KG menu reduced mean selection regret from 0.043153 for expected Uniform fixed to 0.013264 (Table 1). The pre-specified primary paired selected-gain difference was +0.029889, with saved 95% root-bootstrap interval $[0.005868, 0.058099]$ and sign-flip $p = 0.02148$. Expected Uniform averages exactly over all equally likely fixed-size query subsets. The comparison against the separately registered single Uniform draw had interval $[-0.010829, 0.067025]$ and did not establish an advantage. Method-level intervals in Figure 4 do not substitute for these paired contrasts.

Stopping and the HF menu. Early stopping reduced mean HF consumption from 16,384 to 12,083.2 hands, a saving of 4,300.8 hands (26.25%; Figure 1). The saved paired hands-difference interval is $[-6007.4667, -2662.4]$. Observed selected-gain differences were zero in all 30 confirmation roots. This is a query-sample saving in offline replay, not a measured reduction in total bank-generation cost, runtime, or cloud charges, and it is not a guarantee of losslessness on other tasks. The query-size menu did not improve selected gain over fixed-size PIVOT-KG: the menu-minus-fixed difference was -0.013161 with interval $[-0.029576, 0]$. The protocol’s menu version remains the primary method; the stronger fixed-size point estimate does not replace it retrospectively.

Other component findings. Table 2 and Figure 2 retain the null and negative component comparisons. The results do not establish that the menu selector outperforms IVR or LUCB, or that paired queries improve final selected gain over the unpaired variant. Exact versus noisy calibration produced equal selected gains for the observed roots; the noisy-calibration allocation contrast against expected Uniform was positive. Secondary intervals are pointwise and have no multiplicity adjustment. LORO gates C1–C3 passed, with held-out 95% predictive coverage 0.9583 and 0.9688 for short and long adaptation. The confirmation summary’s gate fields are null; the gate evidence comes from the separately archived LORO summary. All statistics below are the previously saved results; this supplement performs no new statistical estimation.

1.1 E1: all seven pre-specified settings

The separate E1 suite used the earlier candidate family and its own protocol-defined Uniform comparator, with 30 confirmation roots per setting and a primary budget of 16,384 HF hands. It must not be pooled with the v4 cohort or interpreted as the same expected-Uniform experiment. Figure 5 retains all seven settings and both the frozen and corrected analyses of the same saved

Table 1: All 13 reported methods at long adaptation and the primary cap, with exact calibration and 30 confirmation roots. Lower regret and higher selected deployment gain are better. All-HF is a four-times-budget reference, while no-HF and proxy-only use zero new HF hands. Uniform fixed is an exact query-subset expectation; Uniform small uses the saved 64-draw Monte Carlo average.

Method	Mean regret	Mean gain	Mean HF hands
PIVOT-KG fixed	0.000104	0.119670	16,384
Top-proxy fixed	0.004017	0.115757	16,384
All-HF fixed (reference)	0.007277	0.112496	65,536
PIVOT-KG menu	0.013264	0.106509	16,384
PIVOT-KG menu + stopping	0.013264	0.106509	12,083.2
LUCB fixed	0.018902	0.100871	16,384
IVR menu	0.021306	0.098467	16,384
IVR fixed	0.021384	0.098389	16,384
Uniform small	0.023825	0.095948	16,384
PIVOT-KG menu, unpaired	0.026767	0.093006	16,384
Uniform fixed (expectation)	0.043153	0.076621	16,384
Calibrated, no HF	0.258765	-0.138991	0
Proxy-only	0.670558	-0.550785	0

worlds. These two analyses are not independent replications. Table 3 reports the corrected contrasts. Pointwise lower bounds are positive for four settings; the $\eta = 0.05$ point estimate is negative, the $\eta = 0.40$ interval touches zero, and the $P(\text{exploiter}) = 0.95$ effect is zero. These secondary intervals are not adjusted for the seven settings; they do not establish a joint family-wide significance claim or a monotonic relation between response strength and PIVOT value.

Table 2: Saved paired component contrasts at the primary cap, $N = 30$. All rows report selected-gain differences except the explicitly labeled HF-hands row. Positive values favor the first term. H_A versus expected Uniform is the pre-specified primary comparison. Remaining intervals are unadjusted; zero-width intervals describe these observed roots only.

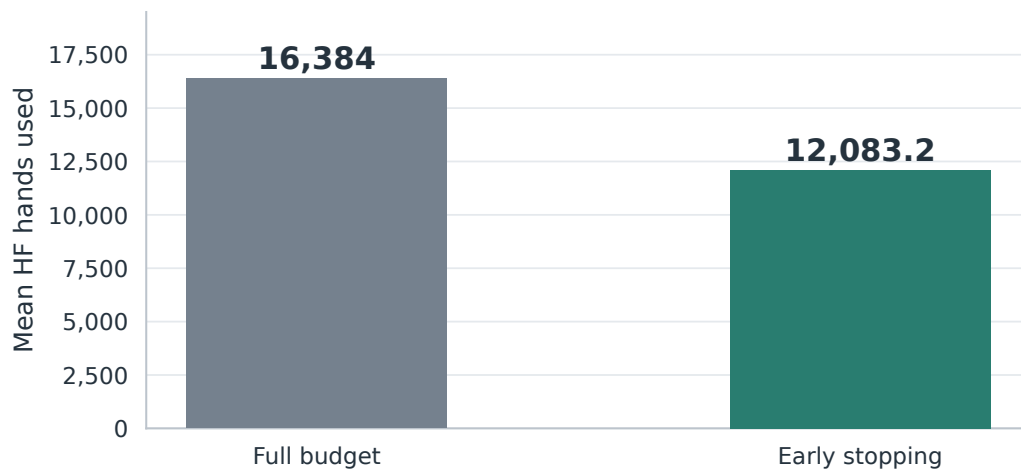
ID	Contrast	Mean difference	95% interval
H_A	Menu – expected Uniform	+0.029889	[0.005868, 0.058099]
H_A	Menu – registered Uniform draw	+0.022629	[−0.010829, 0.067025]
H_B	Menu – IVR menu	+0.008042	[0, 0.021976]
H_{B2}	Menu – LUCB fixed	+0.005638	[−0.024932, 0.049394]
H_{B3}	Menu – top-proxy fixed	−0.009248	[−0.026208, 0.002945]
H_C	Paired – unpaired menu	+0.013503	[−0.001915, 0.032026]
H_D	Exact – noisy calibration	0	[0, 0]
H_D	Noisy menu – expected Uniform	+0.028791	[0.006160, 0.054610]
H_E	Stopping – full menu (gain)	0	[0, 0]
H_E	Stopping – full menu (HF hands)	−4300.8	[−6007.4667, −2662.4]
H_F	Menu – fixed size	−0.013161	[−0.029576, 0]
H_G	Long – short allocation contrast	+0.029889	[0.005868, 0.058099]

Table 3: All seven E1 corrected PIVOT-KG-minus-Uniform selected-gain contrasts. Saved pointwise 95% root-bootstrap intervals, 30 roots per setting. These results are separate from the second-cohort v4 main comparison.

Setting	Mean gain difference	95% interval
$\eta = 0.05$	−0.014819	[−0.032008, 0.000248]
$\eta = 0.10$	+0.033218	[0.006116, 0.066191]
$\eta = 0.25$	+0.014826	[0.002280, 0.031415]
$\eta = 0.40$	+0.001501	[0, 0.004504]
$P(\text{exploiter}) = 0.80$	+0.023550	[0.003317, 0.052706]
$P(\text{exploiter}) = 0.95$	0	[0, 0]
Continuous mixture	+0.051202	[0.003287, 0.112217]

Stopping saves 26.25% of HF hands

Leduc v4 · PIVOT-KG menu · long adaptation · 30 roots

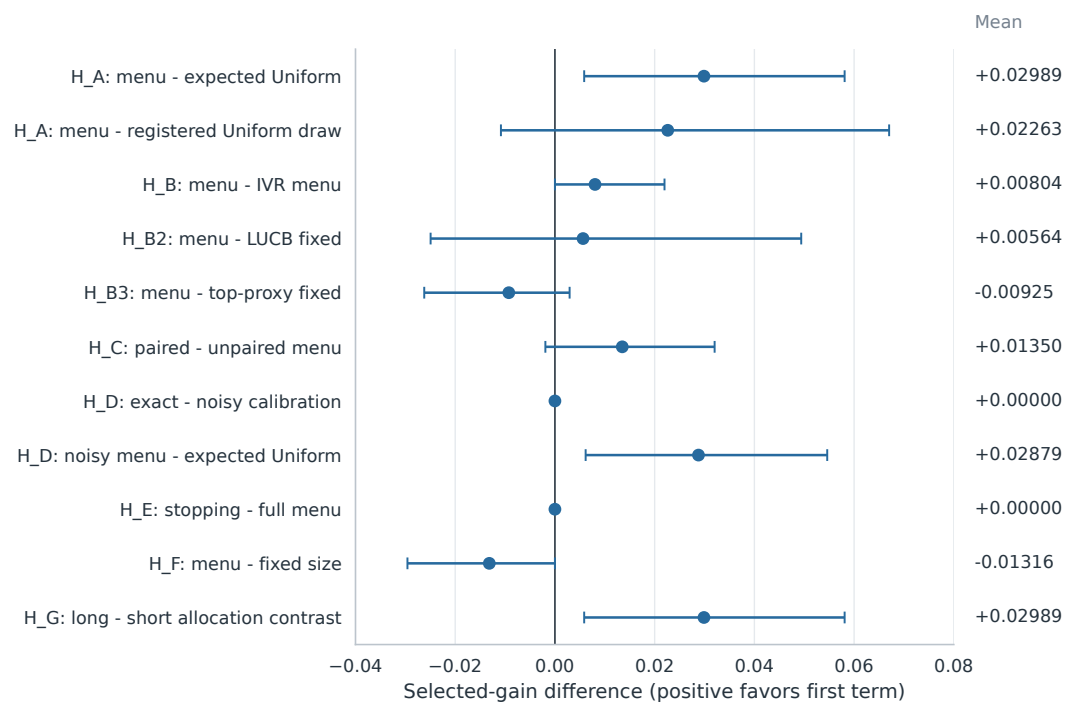


Observed selected-gain difference: 0 in all 30 roots.
Saved mean hands difference: $-4,300.8$; 95% CI $[-6,007.5, -2,662.4]$.
Bars show means; the interval above is for the paired difference.

Figure 1: HF hands consumed by PIVOT-KG menu with and without stopping. Means are shown without invented per-bar intervals; the saved paired difference interval is stated separately. Final selected-gain differences were zero for all 30 observed roots. This measures query-sample use on sealed banks, not end-to-end computational cost.

Component contrasts retain the mixed findings

Leduc v4 · long adaptation · primary budget 16,384 · 30 roots

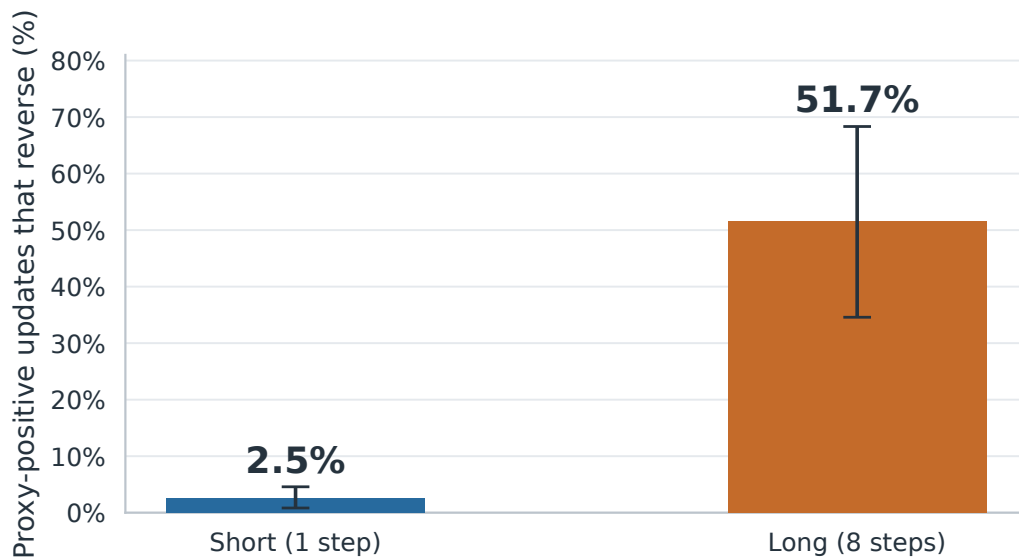


All top-level gain contrasts with saved paired 95% CIs; H_G is the long-minus-short interaction.
H_A expected Uniform is the prespecified primary comparison. Other intervals are unadjusted.
A zero-width interval is the recorded result; it is not a claim of universal equivalence.

Figure 2: All saved top-level gain contrasts, including the menu-versus-fixed null/negative finding. Error bars are saved paired root-bootstrap 95% intervals; secondary comparisons are unadjusted. The HF-hands contrast is reported separately.

Longer adaptation increases reversals

Leduc v4 · 30 confirmation roots · mean and saved 95% CI



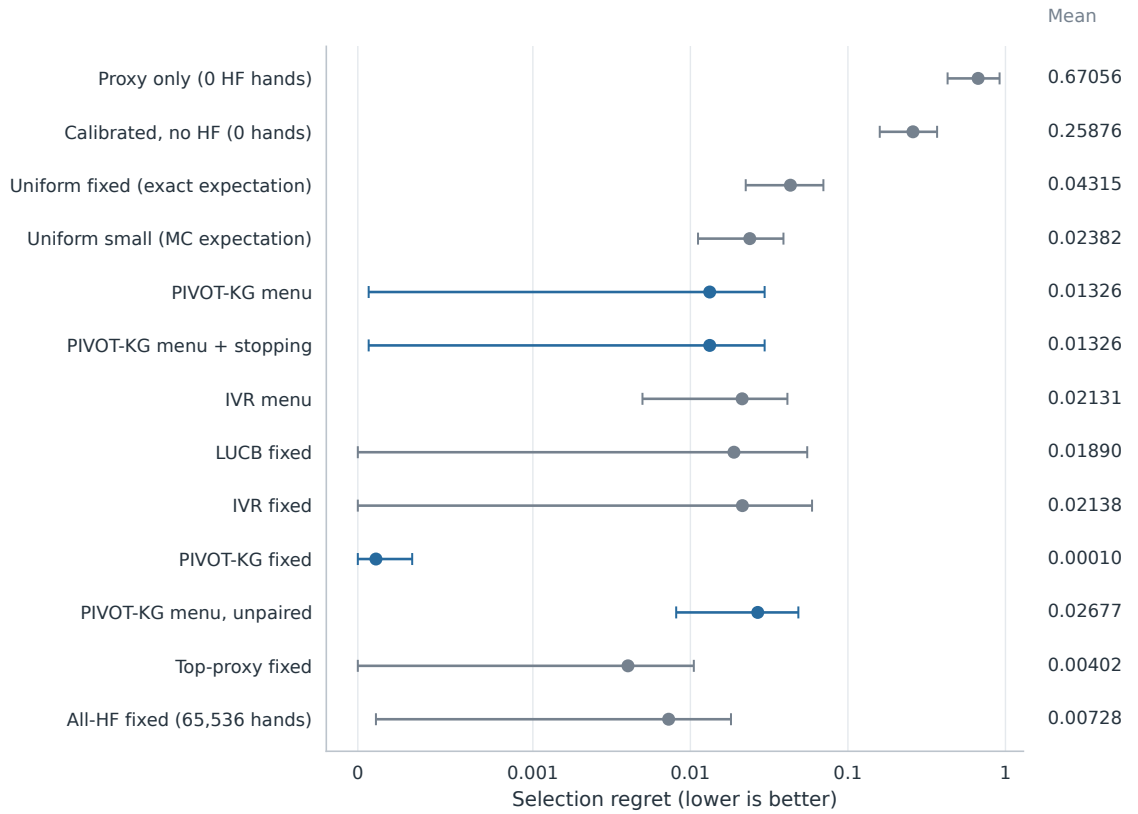
Conditional on positive proxy gain; computed within each root, then averaged. A reversal means deployment gain < 0 .

Figure 3: Proxy-positive improvement reversal in the second Leduc cohort. Within each root, the rate is the fraction of candidates with positive proxy gain whose deployment gain is negative. Bars show the mean of 30 root-level rates and the saved 95% root-bootstrap intervals for 1-step and 8-step adaptation. Candidates within a root are not independent experimental units.

Selection quality across all reported methods

Leduc v4 · long adaptation (8) · exact calibration · 30 roots

All HF methods use 16,384 hands, except stopping (12,083.2 mean) and All-HF (65,536).



Points and intervals are saved root means and 95% bootstrap CIs.

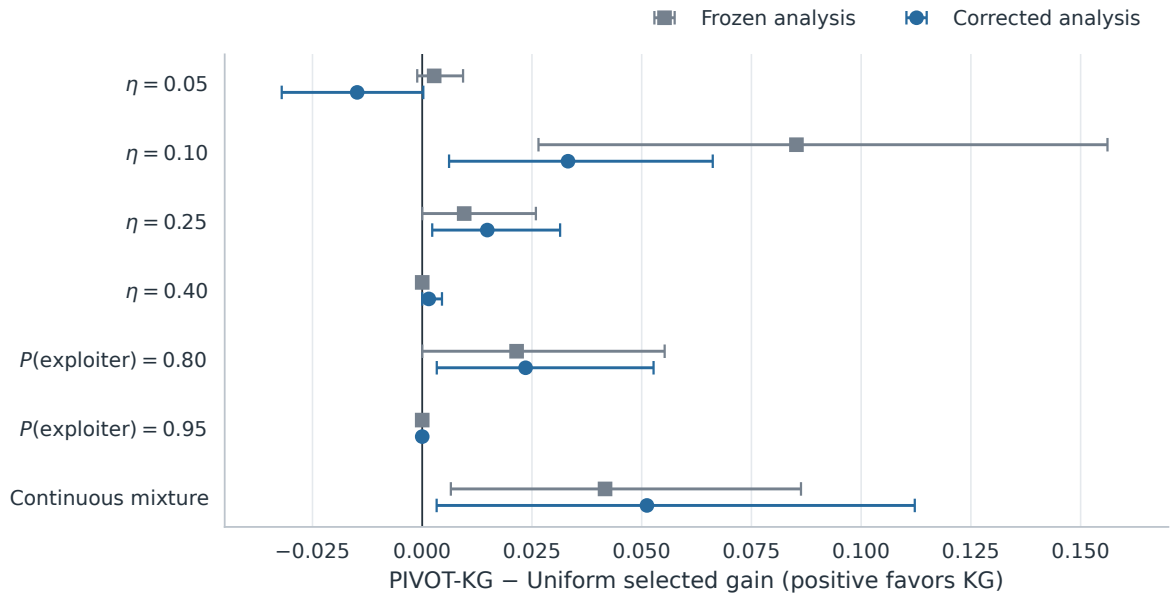
Symmetric-log x axis: linear near zero (threshold 0.001), logarithmic beyond it.

Uniform fixed is the exact query-subset expectation; Uniform small averages 64 Monte Carlo draws.

Figure 4: All 13 reported methods and saved 95% root-bootstrap intervals at long adaptation. Regret is the best attainable deployment gain, including retaining the incumbent, minus the selected gain. Budget exceptions are labeled. The symmetric-log axis is linear within 0.001 of zero; visual distances are not on a globally linear scale. Paired method comparisons are reported in Table 2.

All seven E1 settings, including null and negative results

Leduc v3 suite · long adaptation · primary budget 16,384 hands · 30 roots per setting



Frozen and corrected analyses use the same saved worlds; they are not independent replications. Pointwise saved 95% CIs; no adjustment for seven settings. The effect is not uniformly positive.

Figure 5: All seven E1 settings, including null and negative results. Frozen and corrected analyses use the same saved worlds and differ in the calibration-noise treatment; they are not independent replications. Error bars are the saved pointwise 95% root-bootstrap intervals without multiplicity adjustment. The primary budget is 16,384 HF hands in each setting.

2 MetaDrive: responsive traffic and a null primary method comparison

Setting and scope. We evaluated six prespecified IDM-controller updates in a native MetaDrive $4 \rightarrow 1 \rightarrow 4$ -lane bottleneck with eight vehicles. The focal vehicle is evaluated against seven background vehicles. The proxy fixes the background controller parameters, while deployment selects a shared background-controller profile separately for each candidate and for the incumbent, maximizing the background vehicles’ own mean return. The short and long response conditions search 4 and 12 prespecified profiles using two training scenarios per profile; this is finite controller search, not neural-policy training or an exact best response. The focal update is scored against the incumbent under their respective induced responses. Native cumulative return is the outcome, and greater return alone must not be interpreted as greater driving safety. This is a custom response extension of MetaDrive, distinct from HighwayEnv and from an official adaptive MetaDrive benchmark score.

Protocol and information boundary. The first cohort comprises four pilot roots, 12 calibration roots, and 30 independent confirmation roots. All six candidates are retained, including proxy-negative candidates. Response training, proxy evaluation, HF query observations, and the two independent audit streams use disjoint scenarios. Calibration is frozen before confirmation, and choices are sealed before opening the confirmation audits. The primary condition searches 12 response profiles and permits at most two candidate queries. A query uses four paired candidate/incumbent evaluation scenarios. For response-search size h and positive query budget q , the HF episode charge is $2h + q(2h + 8)$, giving 88 native episodes at the primary condition. The full first cohort generated 36,330 native episodes, including all calibration, query-bank, validation, and audit data; this total experimental cost is separate from a selector’s online budget. There is no HF-menu or adaptive-stopping claim in this cohort.

Primary method result. The prespecified test compares PIVOT-KG with a Uniform allocator using the same calibrated joint posterior and terminal selection rule. Uniform is averaged over 100 allocation draws per root; the independent sample size remains 30, not 3,000. PIVOT-KG attains mean selected deployment gain 9.851, compared with 7.962 for matched-posterior Uniform. The paired difference is +1.889 with 95% root-bootstrap interval $[-1.293, 5.492]$ (10,000 resamples). The interval includes zero, so this first cohort does not establish the primary allocation advantage. All methods at this condition are retained in Table 4, and the paired contrasts are in Table 5. Other budgets, the short-response condition, and comparisons with IVR, LUCB, the Global-VOI heuristic, and zero-HF controls are secondary; all conditions, including null and negative results, remain in the accompanying 72-row method summary and 2,160-row root-level CSV.

Separate author-code comparison. We also call the author E5C implementations of PIVOT-VOI, Uniform, LUCB, and Global-VOI. These are reported separately from PIVOT-KG and the matched-posterior selectors. The author-code PIVOT-VOI minus author-code Uniform contrast is -0.702 with 95% interval $[-6.028, 4.658]$, which also does not establish an advantage. This comparison includes the author methods’ different treatment of unqueried candidates and therefore is not an isolated test of the KG allocation rule. The matched-posterior “Global-VOI heuristic” is not the author Global-VOI implementation.

Response mechanism: post-hoc diagnostics. On the completed 30-root confirmation bank, the long response increases background return relative to fixed controllers by 16.811 [13.263, 20.443]. The mean absolute discrepancy between fixed-background and responding-background update gains is 9.392 for short response and 12.188 for long response; their paired difference is 2.796 [0.945, 4.844] (Table 6). Here the fixed-background reference is measured on matched independent audit scenarios; it is not a re-labelling of the original four-scenario proxy score. Twelve candidate/root pairs

Table 4: MetaDrive first cohort: primary condition (12 response profiles, at most two HF candidate queries), 30 independent confirmation roots. Gain is the selected update minus the incumbent in native cumulative-return units. Intervals are marginal 95% root-bootstrap intervals; the paired contrasts in Table 5 determine the comparison evidence. Zero-HF methods are references. All positive-budget methods here cost 88 native episodes, including response search and paired query evaluation.

Method	Gain	95% CI	HF episodes
Proxy only	6.563	[0.104, 12.772]	0
Calibrated, no HF	8.183	[3.544, 12.801]	0
Uniform (registered draw)	8.626	[3.964, 13.404]	88
Uniform (100-allocation mean)	7.962	[4.124, 11.921]	88
PIVOT-KG	9.851	[5.649, 14.375]	88
IVR	10.047	[5.637, 14.656]	88
LUCB	9.093	[4.838, 13.515]	88
Global-VOI heuristic	8.618	[3.975, 13.346]	88
Author Uniform (E5C)	8.220	[2.594, 13.609]	88
Author PIVOT-VOI (E5C)	7.518	[2.975, 12.063]	88
Author LUCB (E5C)	7.472	[3.403, 11.845]	88
Author Global-VOI (E5C)	7.649	[3.445, 11.877]	88

Table 5: Paired gain contrasts at the primary MetaDrive condition. Both intervals include zero. The author-code comparison is secondary and uses its own Uniform comparator; it must not be conflated with the matched-posterior allocation comparison.

Contrast	Gain difference	95% CI
PIVOT-KG – matched Uniform	+1.889	[−1.293, 5.492]
Author PIVOT-VOI – author Uniform	−0.702	[−6.028, 4.658]

across nine roots have a positive fixed-background gain and a negative long-response gain in both independent audit streams; nine of these pairs also have positive original proxy scores. These diagnostics support the occurrence of response-induced improvement mismatch in this setting. They were computed after cohort completion and do not replace the prespecified primary method test.

Interpretation and cohort boundary. The first cohort supplies evidence for the motivating mechanism while leaving the primary method advantage unresolved. It does not show that a larger response gap necessarily produces larger PIVOT gains, nor that PIVOT is ineffective in general. Absolute regret against a maximum of noisy audit means is not used as the primary endpoint because that maximum has selection bias. This addition contains only the first completed cohort; an independently seeded precision follow-up is a separate cohort and contributes no result here.

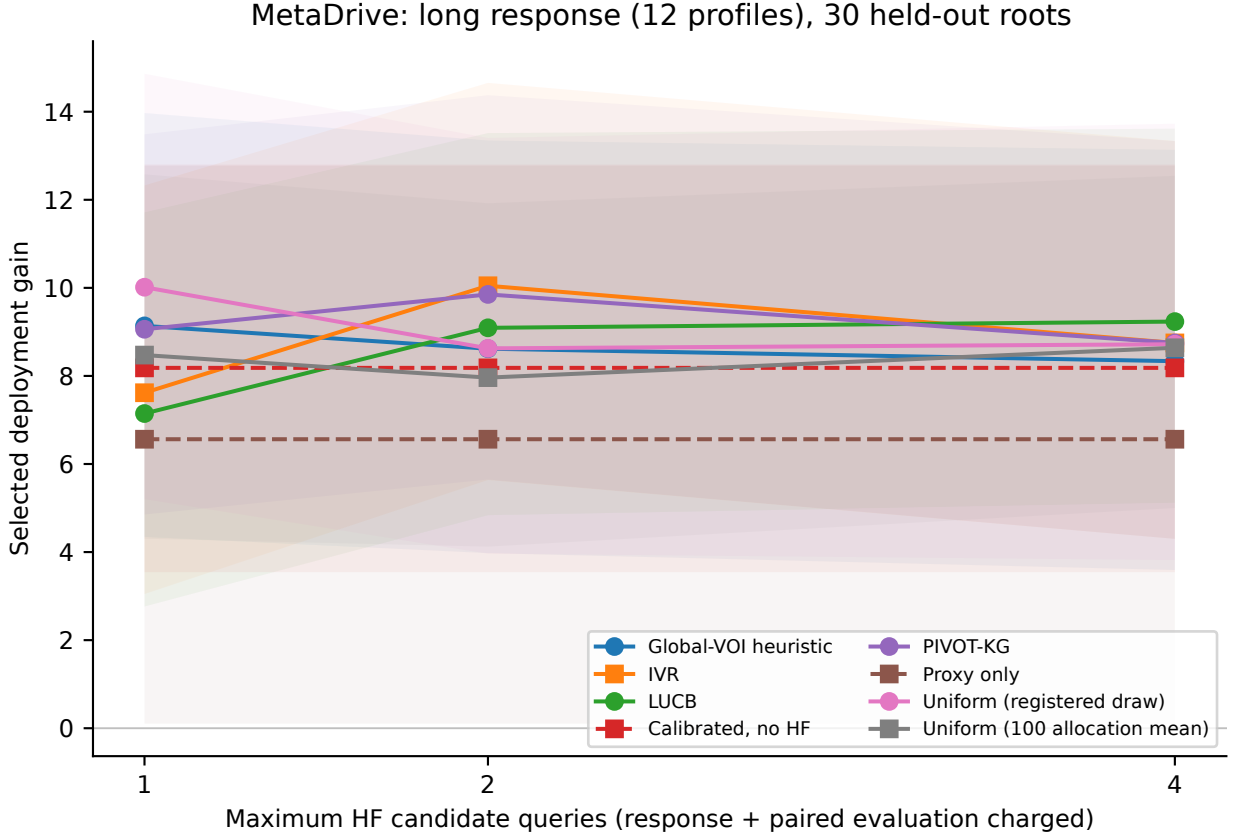


Figure 6: First-cohort MetaDrive confirmation: matched-posterior selectors under the long response condition (12 profiles), 30 independent roots. Higher independent-audit selected gain is better. The prespecified primary budget is two candidate queries. Shading gives marginal 95% root-bootstrap intervals, not the paired contrast interval; significance must not be inferred from band overlap. Proxy-only and calibrated no-HF controls cost zero HF episodes despite being repeated across the query axis.

Table 6: Post-hoc mechanism diagnostics on the 30 confirmation roots. “Short” and “long” search 4 and 12 background-controller profiles, respectively. The gain gap compares an update under fixed and responding backgrounds using matched independent audit scenarios. These diagnostics were computed after the first cohort completed and are not its prespecified primary method test.

Quantity	Mean	95% CI
Background return: long – fixed	+16.811	[13.263, 20.443]
Absolute update-gain gap: short	9.392	[7.819, 10.896]
Absolute update-gain gap: long	12.188	[10.698, 13.699]
Gap: long – short	+2.796	[0.945, 4.844]

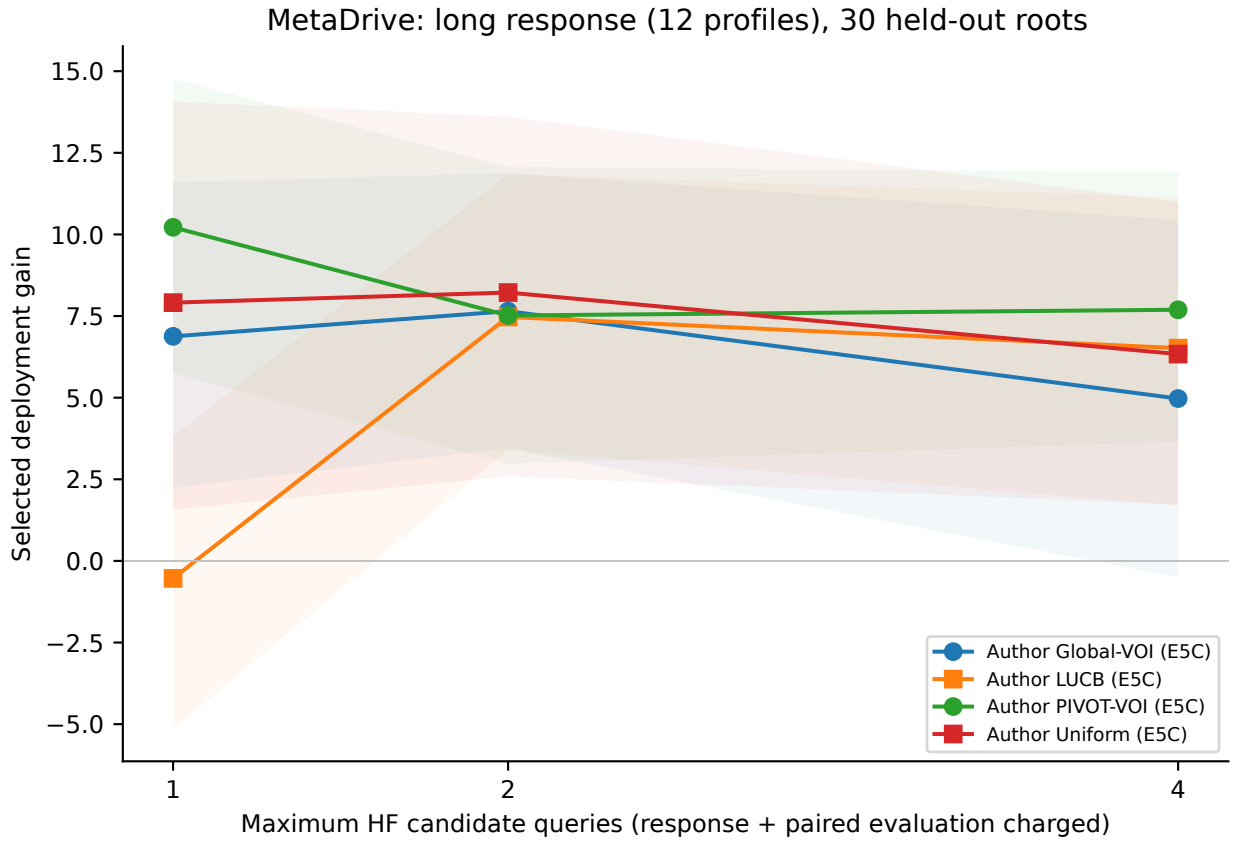


Figure 7: Separate author E5C implementations on the same first-cohort MetaDrive confirmation roots and long response condition. These curves are secondary comparisons. At two queries, the paired PIVOT-VOI minus Uniform interval includes zero. The shaded marginal intervals have the same interpretation as in Figure 6.